

MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 19: Surrogates for Long-Term Treatment Effects

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Application: Estimating Long-Term Returns on Investment

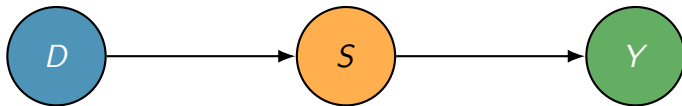
- ▶ Companies frequently deploys new discount or customer support programs
- ▶ Which of these programs (“investments”) are more successful than others?
- ▶ Success is a **long-term** objective: what is the effect of the program on the two-year customer journey (e.g., effect on two-year revenue)
- ▶ We cannot wait two years to evaluate a program
- ▶ **Main Question.** Can we construct estimates of the values of these programs with **short-term** data, e.g. after 6 months?



Battocchi, Dillon, Hei, Lewis, Oprescu & Syrgkanis (2021), *NeurIPS*.

Long-Term Effects from Short-Term Surrogates

- ▶ Suppose that there are many short-term signals S that are indicative of a customer's long-term reward Y (e.g. the next 6-month purchase patterns of a customer could be indicative of their long-term spend)
- ▶ Suppose that investment program D affects long-term rewards if and only if it affects these short-term signals



- ▶ We will call these short-term signals S surrogates

Real-World Applications of Surrogates

Healthcare, Public Policy, and Tech

Application: Job Training Programs (Public Policy)

- ▶ **Setting:** Greater Avenues for Independence (GAIN) job training program in California (1980s) for welfare recipients
- ▶ **Treatment:** Job assistance (search help, education, training)
- ▶ **Long-term outcome:** 9-year post-program earnings
- ▶ **Surrogates:** First 6 quarters of employment rates and earnings
- ▶ **Key finding:** Surrogate index accurately predicted long-term earnings effects across counties, reducing evaluation time from 9 years to ~ 1.5 years

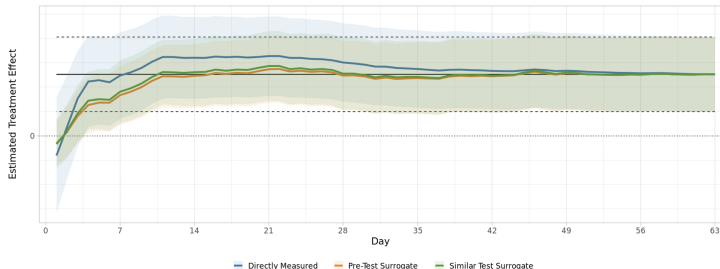


Athey, Chetty, Imbens & Kang (2019/2025), *Review of Economic Studies*

Application: A/B Testing at Netflix (Tech)

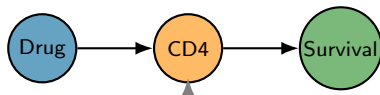
- ▶ **Setting:** 200 A/B tests at Netflix evaluating personalization algorithms
- ▶ **Treatment:** Changes to content recommendation or UI
- ▶ **Long-term outcome:** Day 63 member retention/engagement
- ▶ **Surrogates:** 14-day “auto-surrogate” metrics (short-term observations of the long-term outcome itself)
- ▶ **Key finding:** Decisions using 14-day surrogate index were $\sim 95\%$ consistent with 63-day direct measurement

Zhang, Zhao, Dimakopoulou, Le & Kallus (2023), arXiv:2311.11922



Application: Accelerated Drug Approval (Healthcare)

- ▶ **Setting:** FDA's Accelerated Approval pathway for life-threatening diseases (HIV, cancer, diabetes)
- ▶ **Treatment:** Novel drugs and therapies
- ▶ **Long-term outcome:** Overall survival, major adverse events
- ▶ **Surrogates:** Biomarkers — CD4 count (HIV), tumor shrinkage (cancer), HbA1c (diabetes)
- ▶ **Key finding:** Drugs approved based on surrogate endpoints become available **years earlier**. E.g., antiretroviral therapies approved via CD4 count before long-term survival data was available



Approve based on
surrogate endpoint

Prentice (1989), *Statistics in Medicine*; FDA Accelerated Approval pathway

Identification

Identification of Long Term Effects with Surrogates

Causal Inference with Surrogates 101

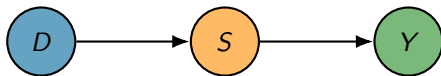
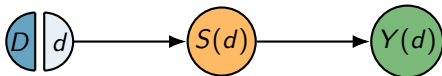
Since long-term effect goes only through surrogates:

(effect on long-term reward) = (effect on forecasted long-term reward based on surrogates)

$$\mathbb{E}[Y(d)] = \mathbb{E}[Y|D=d] \quad (\text{Exogeneity of Treatment})$$

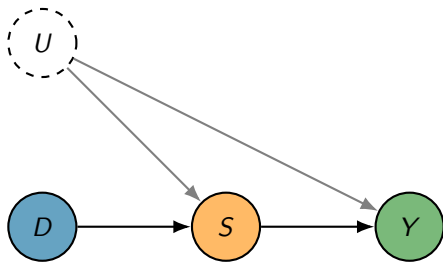
$$= \mathbb{E}[\mathbb{E}[Y|D, S]|D=d] \quad (\text{Tower Law})$$

$$= \mathbb{E}\left[\underbrace{\mathbb{E}[Y|S]}_{\substack{d\text{-separation of } D \text{ from } Y \\ \text{by } S \text{ in DAG}}} \mid D=d\right] = \mathbb{E}\left[\underbrace{h(S)}_{\substack{\text{Forecasted Outcome} \\ \text{from surrogates}}} \mid D=d\right]$$



Caution: Surrogates should not be confounded!

Surrogates are confounded with an unobserved confounder:



(effect on long-term reward) $\neq$ (effect on forecasted long-term reward based on surrogates)

The key property that we used in the previous slide does not hold:

$$Y \perp\!\!\!\perp D \mid S$$

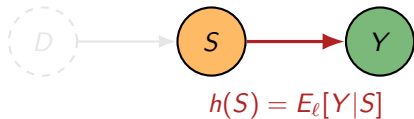
Collider path opens by conditioning! We can block it by conditioning on U (had we observed it)

$$Y \perp\!\!\!\perp D \mid S, U$$

Causal Inference with Surrogates 101

historical/long-term (ℓ)

- ▶ Contains long-term Y and surrogates S
- ▶ Relationship between Y and S the same as in short-term dataset (s)



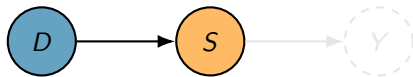
1. Estimate

$$h(S) := \mathbb{E}_{\ell}[Y|S] \quad (\text{surrogate index})$$

from (ℓ) by regressing $Y \sim S$

recent/short-term (s)

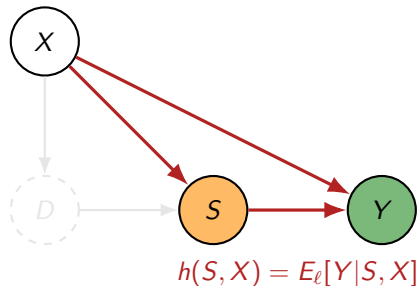
- ▶ Contains treatment D and surrogates S
- ▶ Does not contain long term outcome Y



2. Impute expected long-term outcomes in (s)
3. Regress $h(S) \sim D$ to estimate effect of D on long-term outcome Y from (ℓ)

Causal Inference with Surrogates

long-term data (ℓ)

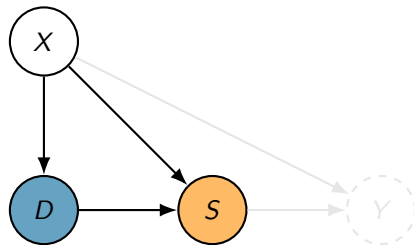


1. Estimate

$$h(S, X) := \mathbb{E}_{\ell}[Y|S, X] \quad (\text{surrogate index})$$

by regressing $Y \sim S, X$

short-term data (s)



2. Impute expected long-term outcomes in (s)
3. Estimate average effect of D on $h(S, X)$ by conditioning on X

Causal Inference with Surrogates 102

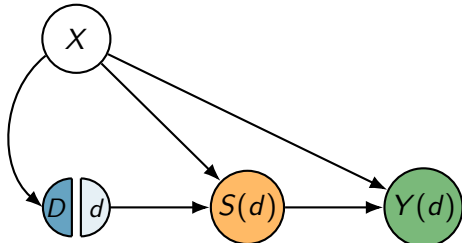
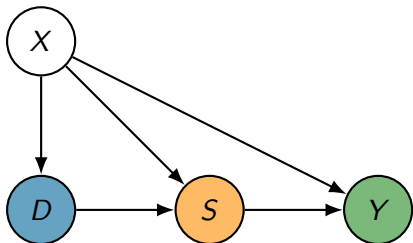
$$\tau(d) := \mathbb{E}_s[Y(d)] = \mathbb{E}_s[\mathbb{E}_s[Y|D = d, X]], \quad (g\text{-formula, } Y(d) \perp\!\!\!\perp D \mid X)$$

- Surrogacy assumption implies $Y \perp\!\!\!\perp D \mid S, X$:

$$\mathbb{E}_s[Y|D = d, X] = \mathbb{E}_s[\mathbb{E}_s[Y|D, S, X] \mid D = d, X] = \mathbb{E}_s[\mathbb{E}_s[Y \mid S, X] \mid D = d, X]$$

- Invariance between short and long term implies

$$\mathbb{E}_s[Y \mid S, X] = \mathbb{E}_\ell[Y \mid S, X] = h(S, X)$$



Nested Regression Estimand

Surrogacy based g-formula

$$\tau(d) = \mathbb{E}_s[g(d, X)]$$

$$g(D, X) = \mathbb{E}_s[h(S, X)|D, X]$$

$$h(S, X) = \mathbb{E}_\ell[Y|S, X]$$

- ▶ Target parameter is averaged over short-term data distribution
- ▶ g regression function is trained on the outputs of the h regression function
- ▶ Surrogate Index Model h is trained on long-term data distribution

Estimation

Doubly Robust Estimation of Long Term Effects with Surrogates

Debiased Insensitive Formula

Surrogacy based g-formula

$$\tau(d) = \mathbb{E}_s[g(d, X)]$$

$$g(D, X) = \mathbb{E}_s[h(S, X) | D, X]$$

$$h(S, X) = \mathbb{E}_\ell[Y | S, X]$$

Debiased formula

$$\tau(d) = E_s \left[g(d, X) + \underbrace{\frac{\mathbf{1}\{D = d\}}{\Pr_s(D = d | X)}}_{\text{Inverse Propensity Weights}} \cdot \underbrace{(h(S, X) - g(D, X))}_{g \text{ regression residual}} \right]$$

Recursive Debiasing

$$\mathbb{E}_s \left[g(d, X) + \underbrace{\frac{\mathbf{1}\{D = d\}}{\Pr_s(D = d|X)}}_{a(D, X)} (h(S, X) - g(D, X)) \right] + \mathbb{E}_\ell \left[w(S, X) \cdot \underbrace{(Y - h(S, X))}_{h \text{ regression residual}} \right]$$

The weighting function $w(S, X)$ is of the form

$$w(S, X) = s(S, X) \cdot \mathbb{E}_s[a(D, X)|S, X] = s(S, X) \cdot \frac{\Pr_s(D = d | S, X)}{\Pr_s(D = d | X)}$$

Surrogate Score s : density ratio of S, X between ℓ and s “environments”

$$s(S, X) = \frac{p(S, X|E = s)}{p(S, X|E = \ell)} = \frac{\Pr(E = s|S, X) \Pr(E = \ell)}{\Pr(E = \ell|S, X) \Pr(E = s)}$$

Doubly Robust Formula for ATE with Surrogates

For the ATE we just take the difference of the two formulas for $d = 1$ and $d = 0$:

$$\text{ATE} = \tau(1) - \tau(0)$$

Which simplifies to:

$$\begin{aligned} \theta = \mathbb{E}_s \left[g(1, X) + g(0, X) + \frac{D - p(X)}{p(X)(1 - p(X))} (h(S, X) - g(D, X)) \right] \\ + \mathbb{E}_\ell \left[s(S, X) \left(\frac{\pi(S, X)}{p(X)} - \frac{1 - \pi(S, X)}{1 - p(X)} \right) \cdot (Y - h(S, X)) \right] \end{aligned}$$

where

$$p(X) = \Pr_s(D = 1 \mid X)$$

$$\pi(S, X) = \Pr_s(D = 1 \mid S, X)$$

DR Surrogate ATE Estimator: Pseudocode — Nuisance Estimation

```
def dr_surrogate_ate(data_s, data_l, S_cols, X_cols, K=5, # ATE = E[Y(1)]-E[Y(0)]
                    reg_model, clf_model):
    SX, DX = S_cols + X_cols, ['D'] + X_cols
    n_s, n_l = len(data_s), len(data_l)
    h_hat_l, h_hat_s = np.zeros(n_l), np.zeros(n_s)
    g_hat, g_l_hat, g0_hat = np.zeros(n_s), np.zeros(n_s), np.zeros(n_s)
    pi_hat, w1_hat, w0_hat = np.zeros(n_s), np.zeros(n_l), np.zeros(n_l)
    data_s['E'], data_l['E'] = 1, 0
    data_all = pd.concat([data_l, data_s]) # first n_l long-term, then n_s short-term
    for train, test in KFold(n_splits=K, shuffle=True).split(data_all):
        tr_l, te_l = train[train < n_l], test[test < n_l]
        tr_s, te_s = train[train >= n_l] - n_l, test[test >= n_l] - n_l
        # h=E_l[Y|S,X]
        h_m = clone(reg_model).fit(data_l.iloc[tr_l][SX], data_l.iloc[tr_l]['Y'])
        h_hat_l[te_l] = h_m.predict(data_l.iloc[te_l][SX])
        h_hat_s[te_s] = h_m.predict(data_s.iloc[te_s][SX])
        # g=E_s[h|D,X]
        g_m = clone(reg_model).fit(data_s.iloc[tr_s][DX], h_m.predict(data_s.iloc[tr_s][SX]))
        g_hat[te_s] = g_m.predict(data_s.iloc[te_s][DX])
        X_te = data_s.iloc[te_s][X_cols]
        te_1 = X_te.copy(); te_1.insert(0, 'D', 1); te_0 = X_te.copy(); te_0.insert(0, 'D', 0)
        g1_hat[te_s], g0_hat[te_s] = g_m.predict(te_1), g_m.predict(te_0)
        # Pr(D|X)
        pi_m = clone(clf_model).fit(data_s.iloc[tr_s][X_cols], data_s.iloc[tr_s]['D'])
        pi_hat[te_s] = pi_m.predict_proba(data_s.iloc[te_s][X_cols])[:, 1]
        # density ratio s(S,X)
        idx_all = np.r_[tr_l, tr_s + n_l]
        s_m = clone(clf_model).fit(data_all.iloc[idx_all][SX], data_all.iloc[idx_all]['E'])
        s_p = s_m.predict_proba(data_l.iloc[te_l][SX])
        s_ratio = (s_p[:, 1] / s_p[:, 0]) * (n_l / n_s)
        # Pr(D|S,X)
        a_m = clone(clf_model).fit(data_s.iloc[tr_s][SX], data_s.iloc[tr_s]['D'])
        a_p = a_m.predict_proba(data_l.iloc[te_l][SX])
        pi_l = pi_m.predict_proba(data_l.iloc[te_l][X_cols])[:, 1]
        w1_hat[te_l] = s_ratio * a_p[:, 1] / pi_l # w_1(S,X)
        w0_hat[te_l] = s_ratio * a_p[:, 0] / (1 - pi_l) # w_0(S,X)
```

DR Surrogate ATE Estimator: Pseudocode — Estimation & Inference

```
# (continued) Difference the two DR formulas and simplify:
# tau = E_s[g(1,X)-g(0,X)] + E_s[ipw*(h(S,X)-g(D,X))] + E_l[(w1-w0)*(Y-h(S,X))]
# where ipw = (D - pi(X)) / (pi(X)*(1-pi(X))) [simplification of 1{D=1}/pi - 1{D=0}/(1-pi)]
D_s, Y_l = data_s['D'].values, data_l['Y'].values
ipw = (D_s - pi_hat) / (pi_hat * (1 - pi_hat)) # simplified IPW for ATE
term1 = np.mean(g1_hat - g0_hat) # plug-in: E[g(1,X) - g(0,X)]
term2 = np.mean(ipw * (h_hat_s - g_hat)) # debiasing for g
term3 = np.mean((w1_hat - w0_hat) * (Y_l - h_hat_l)) # debiasing for h
tau = term1 + term2 + term3
# Influence function for ATE
psi_s = (g1_hat - g0_hat) + ipw * (h_hat_s - g_hat) - tau # n_s-vector
psi_l = (w1_hat - w0_hat) * (Y_l - h_hat_l) # n_l-vector
# Asymptotic variance: Var(tau) = Var(psi_s)/n_s + Var(psi_l)/n_l
se = np.sqrt(np.var(psi_s) / n_s + np.var(psi_l) / n_l)
return tau, se, tau - 1.96 * se, tau + 1.96 * se
```

References I

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